

Algebra: Solving Radical Equations



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DIRECTIONS

Isolate the radical, then square both sides to eliminate it. Solve for the variable and always check for extraneous solutions.

1. Solve: $\sqrt{x} = 5$ Answer: _____	2. Solve: $\sqrt{x} = 9$ Answer: _____
3. Solve: $\sqrt{(x + 3)} = 4$ Answer: _____	4. Solve: $\sqrt{(2x)} = 6$ Answer: _____
5. Solve: $\sqrt{(x - 1)} = 3$ Answer: _____	6. Solve: $\sqrt{(3x + 4)} = 5$ Answer: _____
7. Solve: $2\sqrt{x} = 8$ Answer: _____	8. Solve and check: $\sqrt{(x + 5)} = x - 1$ Answer: _____
9. Solve and check: $\sqrt{(2x - 3)} = x - 3$ Answer: _____	10. Solve: $\sqrt{(5x - 4)} - 3 = 0$ Answer: _____

Based on the Numberbender lesson "How to Find the Solutions of Radical Equations" • youtu.be/lgd7DJ4E5rI

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Answer Key & Solutions

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TEACHER NOTES

Items 1–4: isolate the radical then square — straightforward. Items 5–7: require an algebraic step before squaring. Items 8–9: quadratic results after squaring — check both roots for extraneous solutions. Item 10: two-step isolation.

1. Solve: $\sqrt{x} = 5$

Answer: $x = 25$

Square both sides: $x = 5^2 = 25$.

2. Solve: $\sqrt{x} = 9$

Answer: $x = 81$

Square both sides: $x = 9^2 = 81$.

3. Solve: $\sqrt{x + 3} = 4$

Answer: $x = 13$

Square: $x + 3 = 16$; $x = 13$.

4. Solve: $\sqrt{2x} = 6$

Answer: $x = 18$

Square: $2x = 36$; $x = 18$.

5. Solve: $\sqrt{x - 1} = 3$

Answer: $x = 10$

Square: $x - 1 = 9$; $x = 10$.

6. Solve: $\sqrt{3x + 4} = 5$

Answer: $x = 7$

Square: $3x + 4 = 25$; $3x = 21$; $x = 7$.

7. Solve: $2\sqrt{x} = 8$

Answer: $x = 16$

Divide by 2: $\sqrt{x} = 4$; square: $x = 16$.

8. Solve and check: $\sqrt{x + 5} = x - 1$

Answer: $x = 4$

Square: $x + 5 = (x - 1)^2$; $x^2 - 3x - 4 = 0$; $x = 4$ ✓ ($x = -1$ extraneous).

9. Solve and check: $\sqrt{2x - 3} = x - 3$

Answer: $x = 6$

Square: $2x - 3 = (x - 3)^2$; $x^2 - 8x + 12 = 0$; $x = 6$ ✓ ($x = 2$ extraneous).

10. Solve: $\sqrt{5x - 4} - 3 = 0$

Answer: $x = 13/5$

Add 3: $\sqrt{5x - 4} = 3$; square: $5x - 4 = 9$; $5x = 13$; $x = 13/5$.

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